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🎓 [Google Scholar](#)

Profile

I am a third-year PhD student in Mechanical Engineering at Purdue University, working on predictive control and reinforcement learning (RL) algorithms with applications to buildings. I also explore broader machine learning (ML) methods for AI-enabled applications across domains.

Education

Purdue University, USA PhD in Mechanical Engineering — GPA: 3.89/4.0 Minor in Computational Science and Engineering	2023–2027
University of British Columbia, Canada MS in Mechanical Engineering — GPA: 4.0/4.0 (94%)	2021–2023
University of Tabriz, Iran BS in Mechanical Engineering — GPA: 4.0/4.0 (19.12/20) — Rank: 1/155+	2016–2020

Experience

Applied ML Intern, Tesla, Inc., Sparks, NV, USA • Built supervised ML models (XGBoost, DNNs, SVR, and ensembles) for defect detection and quality assessment. • Developed data pipelines in MySQL and engineered features to extract high-signal inputs for ML models. • Contributed to deployment of a thermal imaging system for non-destructive inspection in drive-unit manufacturing. • Collaborated on projects projected to reduce testing costs by over \$1M annually through improved quality assessment.	05/2025 – 08/2025
Graduate Research Assistant, Purdue University, West Lafayette, IN, USA • Formulated a multi-zone HVAC M&V method to remove savings overestimation from cross-zone heat transfer. • Developed an RL controller for residential HVAC using the OCHRE, saving 17.5% energy while maintaining comfort. • Reviewed 100+ publications on MPC and RL in HVAC, leading to a published review paper in Applied Energy. • Created real-time dashboards for energy/comfort and ML models for occupancy detection (98% accuracy). • Supported commissioning of Herrick Labs by upgrading the building automation system and integrating Modbus. • Designed HVAC control strategies with MPC and DeePC, showing DeePC matched MPC without a system model. • Assisted in developing active current-limiting control for appliances to prevent breaker overloads in electrified homes.	08/2023 – 08/2027
Graduate Research Assistant, University of British Columbia, BC, Canada • Modeled hydrogen injection into natural gas pipelines with CFD by developing ideal- and real-gas mixture models. • Produced a journal paper, three conference papers, two awards, and a master's thesis based on the research findings.	05/2021 – 08/2023
Student Researcher, FortisBC Energy Inc., Surrey, BC, Canada • Performed assessments on hydrogen injection into natural gas pipelines and renewable gas projects (UBC H ₂ Lab). • Worked with engineers and research teams (materials, sensors, AI, safety) to assess industry impacts.	05/2021 – 08/2023

Selected Publications — Full list on [Google Scholar](#)

- A.J. Khabbazi** and K.J. Kircher. Small HVAC control demonstrations in larger buildings often overestimate savings. *American Control Conference (ACC)*, 2026.
- A.J. Khabbazi**, E.N. Pergantis, L.D. Reyes Premer, P. Papageorgiou, A.H. Lee, J.E. Braun, G.P. Henze and K.J. Kircher. Lessons learned from field demonstrations of model predictive control and reinforcement learning for residential and commercial HVAC: A review. *Applied Energy* (2025).
- A.J. Khabbazi**, M. Zabihi, R. Li, M. Hill, V. Chou and J. Quinn. Mixing hydrogen into natural gas distribution pipeline system through Tee junctions. *International Journal of Hydrogen Energy* (2024).
- Additional peer-reviewed conference papers (BuildSys, CISBAT, IHPB, CSME, IGEC; 2021–2025).

Skills

- Expertise:** Predictive Control, Reinforcement Learning, Convex Optimization, Control Theory, Machine Learning, Deep Learning, Generative Models, Industrial IoT, Buildings & HVAC, Thermodynamics
- Graduate Coursework:** Artificial Intelligence, Reinforcement Learning & Control, Optimal Control & Estimation, Convex Optimization, Statistical Methods, Industrial Internet of Things, Distributed Energy Resources, Analysis of Thermal Systems, Advanced Thermodynamics
- Programming & Tools:** Python, MATLAB, C, Git, Linux, Docker
- ML & Optimization:** PyTorch, TensorFlow, scikit-learn, XGBoost, Stable-Baselines3, CVX, CVXPY, Gurobi
- Data Analysis & Visualization:** Pandas, NumPy, SciPy, Matplotlib, Seaborn, Grafana
- Simulation & Testbeds:** Gymnasium, OCHRE, BOPTEST
- Databases & Integration:** MySQL, InfluxDB, Tridium Niagara, Modbus, IO-Link, MTConnect

Selected Honors & Awards

Best PhD Forum Presentation Runner-up Award at ACM BuildSys (2025) — ASHRAE Graduate Student Grant-in-Aid Award (2025) — Best Paper Award at CSME (2023) — Best Presentation Award at CSME (2022) — UBC Graduate Scholarship (2022) — UBC Dean's Entrance Scholarship (2021)